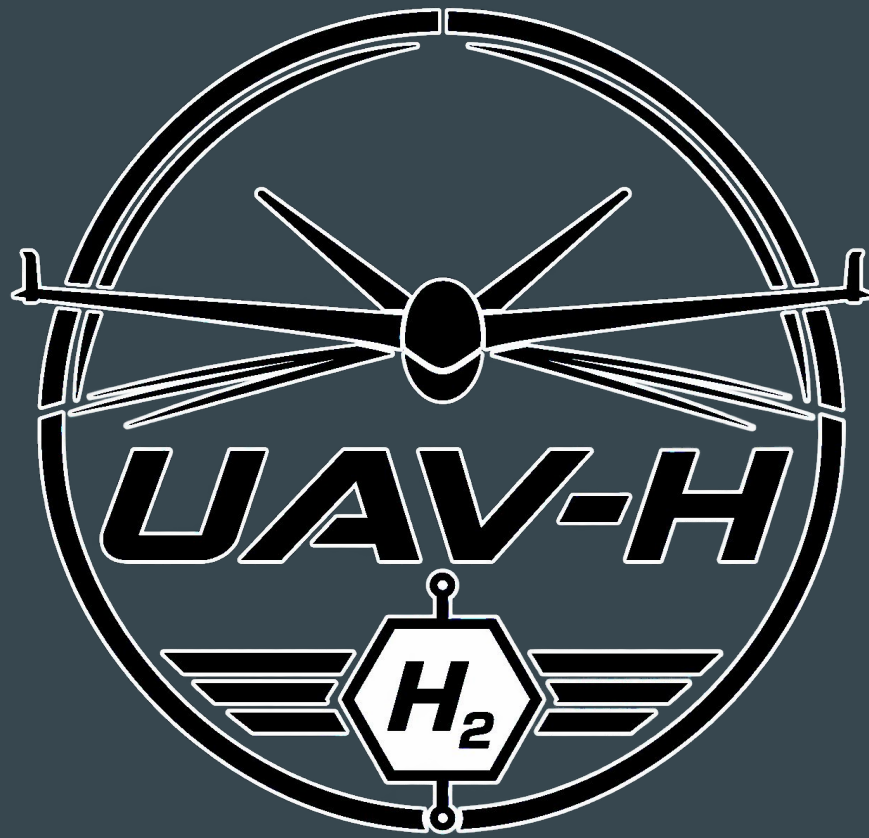




Project Overview

The Team



James Belmar

Sophomore

Chemical Engineering



Georgie Naegele

Sophomore

Mechanical Engineering



Trey Harris

Sophomore

Mechanical Engineering



Bobby Schneider

Sophomore

Aerospace Engineering



We are engineering students at University of Illinois Urbana-Champaign. UAV-H is a part of ACM SIGRobotics, a technical registered student organization on campus.



Our Idea - Build and test a hydrogen-powered UAV focused on endurance.

Why Hydrogen

- It has an energy density 4 times that of lithium ion batteries – decreasing total weight and increasing fuel capacity resulting in a much longer flight times
- It's completely green – the only products are current, water and heat
- It can be “recharged” in minutes – hydrogen gas just needs to be pumped into a tank

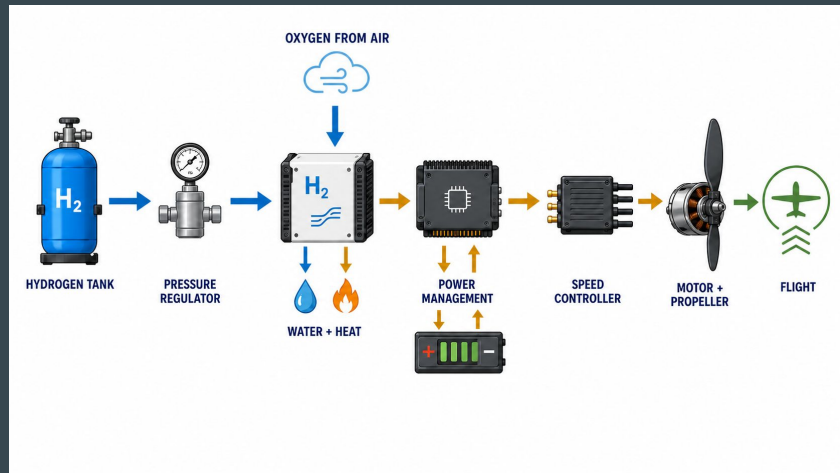
How

- Using hydrogen as fuel in a stack of fuel cells – reacting hydrogen with oxygen to produce water and a usable current

Flow and Function

The system converts stored hydrogen into controlled electrical power and then into flight:

- The regulator safely reduces pressure from the hydrogen tank to an appropriate rate for the fuel cell.
- The fuel cell combines hydrogen and oxygen to generate electricity, water and heat.
- Power-management electronics distribute and regulate the electrical output, as well as receive flight signals.
- A buffer battery provides extra power during takeoff, climbing, and sudden maneuvers.
- The speed controller delivers the required power to the motor and flight controls.
- The motors convert electricity into thrust/movement, producing forward motion, lift, and wing control.



Projected Specs/Goals

- A 100 W hydrogen fuel cell
- 0.1 lb (45 g) compressed hydrogen
- A small lithium battery
- Total assembly mass: 6-10 lbs
- 50+ miles of consecutive flight

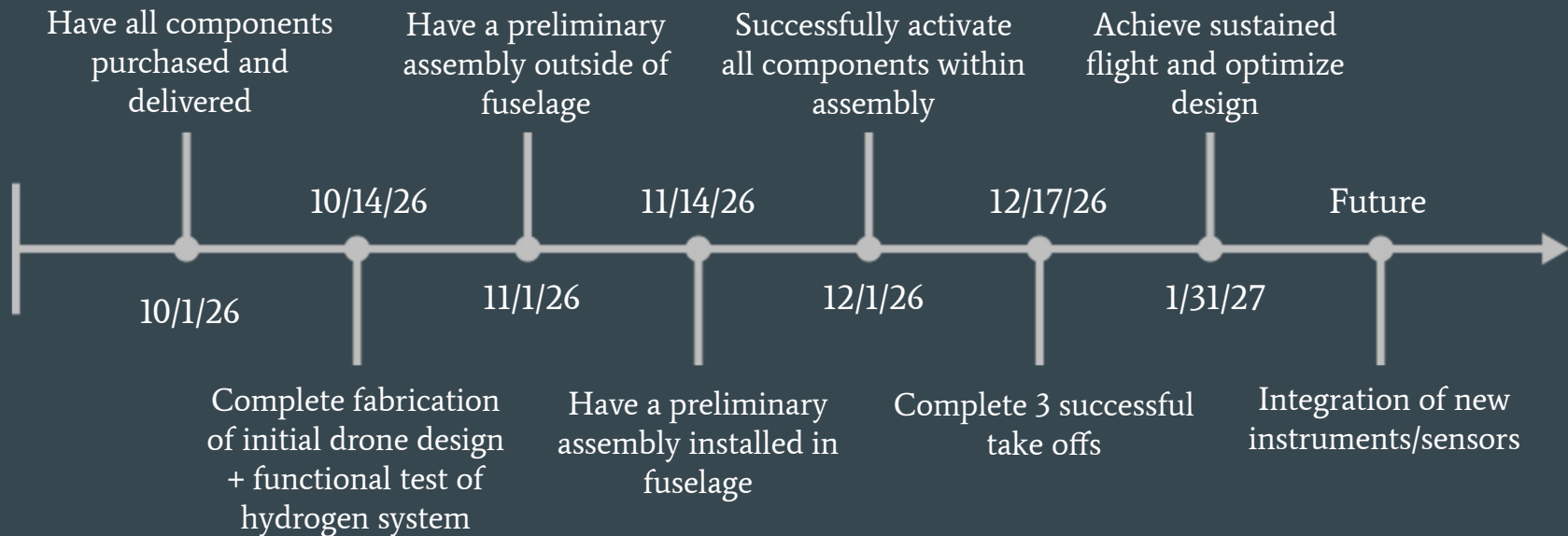
$$R = \frac{V}{c_t} \left(\frac{L}{D} \right) \ln \left(\frac{W_i}{W_f} \right)$$

Breguet range equation

Justification

- Projected UAV specs were derived from existing fuel-cell powered UAVs and the Breguet range equation
- In 2005, the US Naval Research Laboratory flew a 5.6-pound UAV with a 100 watt hydrogen fuel cell for 3 hours and 19 minutes
- The Breguet range equation helped determine our projected flight distance

Project Timeline



*Please note this is a speculative timeline and will be subject to change depending on unforeseen challenges and inevitable iteration

Why Partner With Us?

Brand exposure — Logo/company name on all associated components, published reports, demo footage, and social/SIG Robotics channels

Involvement in green energy technology — Hydrogen aviation is still a growing space; being attached to a credible, data-driven student project signals the company is committed to forging a cleaner future.

Campus & organizational ties — Connection to the UIUC engineering community and ACM, an organization with over 100,000 members nationwide, creates opportunities for future recruiting events, guest talks, or co-sponsored activities beyond this project.

We are open to any suggestions of ways to mutually benefit each other!